

> LAB 1| 22/01/26

↳ 8 cells hidden

> LAB 2| 29/01/26

↳ 7 cells hidden

✓ LAB 3| 05/02/26

```
import numpy as np

x1, x2 = 0.6, 0.1
y = 1

w1, w2, b = 0.2, -0.3, 0.4
lr = 0.001

def sigmoid(z):
    return 1/(1+np.exp(-z))

z = w1*x1 + w2*x2 + b
y_pred = sigmoid(z)

dL_dy_pred = y_pred - y

Dy_pred_dz = y_pred*(1-y_pred)

dL_dz = dL_dy_pred * Dy_pred_dz

dL_dw1 = dL_dz * x1
dL_dw2 = dL_dz * x2
dL_db = dL_dz

w1 -= lr*dL_dw1
w2 -= lr*dL_dw2
b -= lr*dL_db

print(w1, w2, b)

print("Prem Massand,431")
print("Date:05/02/2026")
```

0.20005369592979536 -0.29999105067836745 0.4000894932163256
Prem Massand,431
Date:05/02/2026

```
import numpy as np

# AND data
X = np.array([
    [0,0],
    [0,1],
    [1,0],
    [1,1]
])

y = np.array([0,0,0,1])

w = np.array([0.1, 0.2])
b = 0.3
lr = 0.1

def sigmoid(z):
    return 1/(1+np.exp(-z))

for epoch in range(1000):
    for i in range(len(X)):
        z = np.dot(X[i], w) + b
```

```

y_pred = sigmoid(z)

dz = y_pred - y[i]
dw = dz * X[i]
db = dz
w -= lr * dw
b -= lr * db

print("Weights:", w)
print("Bias:", b)

```

```

print("Prem Massand,431")
print("Date:05/02/2026")

```

```

Weights: [5.60154076 5.59544869]
Bias: -8.565987677634602
Prem Massand,431
Date:05/02/2026

```

```

import numpy as np

# OR data
X = np.array([
    [0,0],
    [0,1],
    [1,0],
    [1,1]
])

y = np.array([0, 1, 1, 1])

w = np.array([0.1, 0.2])
b = 0.3
lr = 0.1

def sigmoid(z):
    return 1/(1+np.exp(-z))

for epoch in range(1000):
    for i in range(len(X)):
        z = np.dot(X[i], w) + b
        y_pred = sigmoid(z)

        dz = y_pred - y[i]
        dw = dz * X[i]
        db = dz

        w -= lr * dw
        b -= lr * db

    print("Weights:", w)
    print("Bias:", b)

print("Prem Massand,431")
print("Date:05/02/2026")

```

```

Weights: [6.7832796 6.78742098]
Bias: -2.9139465005185365
Prem Massand,431
Date:05/02/2026

```

```

import numpy as np
import matplotlib.pyplot as plt

# Inputs
x1, x2 = 0.6, 0.1
y = 1

w1, w2, b = 0.2, -0.3, 0.4
lr = 0.1

def sigmoid(z):
    return 1/(1+np.exp(-z))

loss_list = []

epochs = 1000

```

```

for epoch in range(epochs):

    z = w1*x1 + w2*x2 + b
    y_pred = sigmoid(z)

    loss = -(y*np.log(y_pred) + (1-y)*np.log(1-y_pred))
    loss_list.append(loss)

    dL_dy_pred = y_pred - y
    Dy_pred_dz = y_pred * (1 - y_pred)
    dL_dz = dL_dy_pred * Dy_pred_dz

    dL_dw1 = dL_dz * x1
    dL_dw2 = dL_dz * x2
    dL_db = dL_dz

    # Update
    w1 -= lr * dL_dw1
    w2 -= lr * dL_dw2
    b -= lr * dL_db

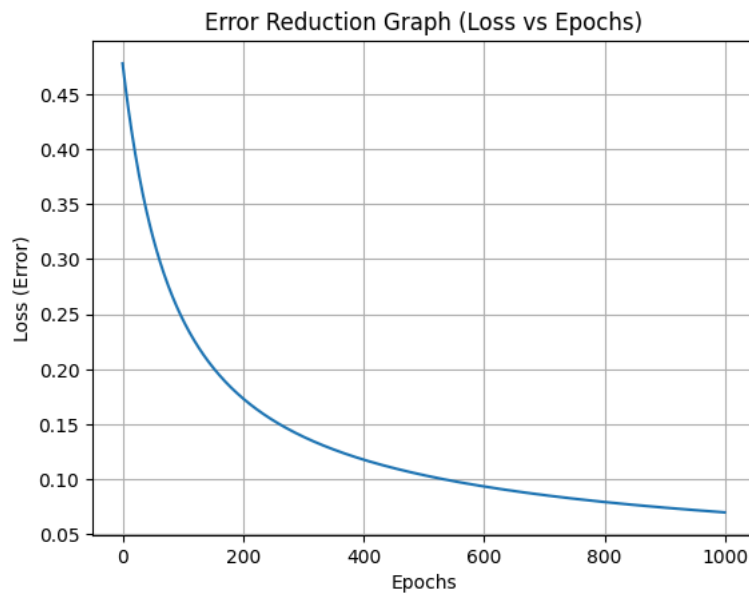
print("Final w1:", w1)
print("Final w2:", w2)
print("Final b :", b)

plt.plot(loss_list)
plt.xlabel("Epochs")
plt.ylabel("Loss (Error)")
plt.title("Error Reduction Graph (Loss vs Epochs)")
plt.grid(True)
plt.show()

print("Prem Massand,431")
print("Date:05/02/2026")

```

Final w1: 1.134656047343927
 Final w2: -0.14422399210934486
 Final b : 1.9577600789065457



Prem Massand,431
 Date:05/02/2026

```

import numpy as np
import matplotlib.pyplot as plt

# Inputs
x1, x2 = 0.6, 0.1
y = 1

def sigmoid(z):
    return 1/(1+np.exp(-z))

learning_rates = [0.001, 0.01, 0.1, 1]

epochs = 1000

```

```

plt.figure(figsize=(10, 6))

for lr in learning_rates:

    w1, w2, b = 0.2, -0.3, 0.4

    loss_list = []

    for epoch in range(epochs):

        z = w1*x1 + w2*x2 + b
        y_pred = sigmoid(z)

        loss = -(y*np.log(y_pred) + (1-y)*np.log(1-y_pred))
        loss_list.append(loss)

        dL_dy_pred = y_pred - y
        Dy_pred_dz = y_pred * (1 - y_pred)
        dL_dz = dL_dy_pred * Dy_pred_dz

        dL_dw1 = dL_dz * x1
        dL_dw2 = dL_dz * x2
        dL_db = dL_dz

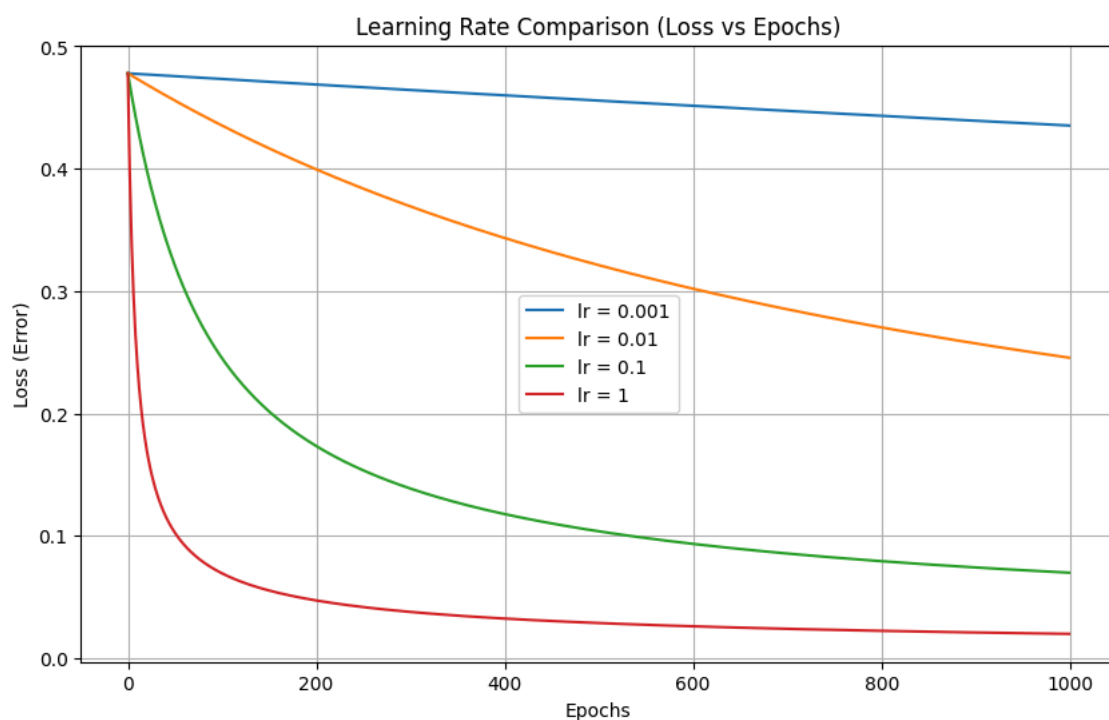
        w1 -= lr * dL_dw1
        w2 -= lr * dL_dw2
        b -= lr * dL_db

    plt.plot(loss_list, label=f"lr = {lr}")

plt.xlabel("Epochs")
plt.ylabel("Loss (Error)")
plt.title("Learning Rate Comparison (Loss vs Epochs)")
plt.grid(True)
plt.legend()
plt.show()

print("Prem Massand,431")
print("Date:05/02/2026")

```



Prem Massand,431
Date:05/02/2026

```

import numpy as np
import matplotlib.pyplot as plt

```

```

# AND data
X = np.array([
    [0,0],
    [0,1],
    [1,0],
    [1,1]
])

y = np.array([0,0,0,1])

def sigmoid(z):
    return 1/(1+np.exp(-z))

learning_rates = [0.001, 0.01, 0.1, 1]

epochs = 1000

plt.figure(figsize=(10, 6))

for lr in learning_rates:

    w = np.array([0.1, 0.2], dtype=float)
    b = 0.3

    loss_list = []

    for epoch in range(epochs):

        total_loss = 0

        for i in range(len(X)):

            z = np.dot(X[i], w) + b
            y_pred = sigmoid(z)

            loss = -(y[i]*np.log(y_pred) + (1-y[i])*np.log(1-y_pred))
            total_loss += loss

            dz = y_pred - y[i]
            dw = dz * X[i]
            db = dz

            w -= lr * dw
            b -= lr * db

        loss_list.append(total_loss / len(X))

    plt.plot(loss_list, label=f"lr = {lr}")

plt.xlabel("Epochs")
plt.ylabel("Average Loss (Error)")
plt.title("AND Gate: Learning Rate Comparison (Loss vs Epochs)")
plt.grid(True)
plt.legend()
plt.show()

lr = 0.1
w = np.array([0.1, 0.2], dtype=float)
b = 0.3

for epoch in range(1000):
    for i in range(len(X)):
        z = np.dot(X[i], w) + b
        y_pred = sigmoid(z)

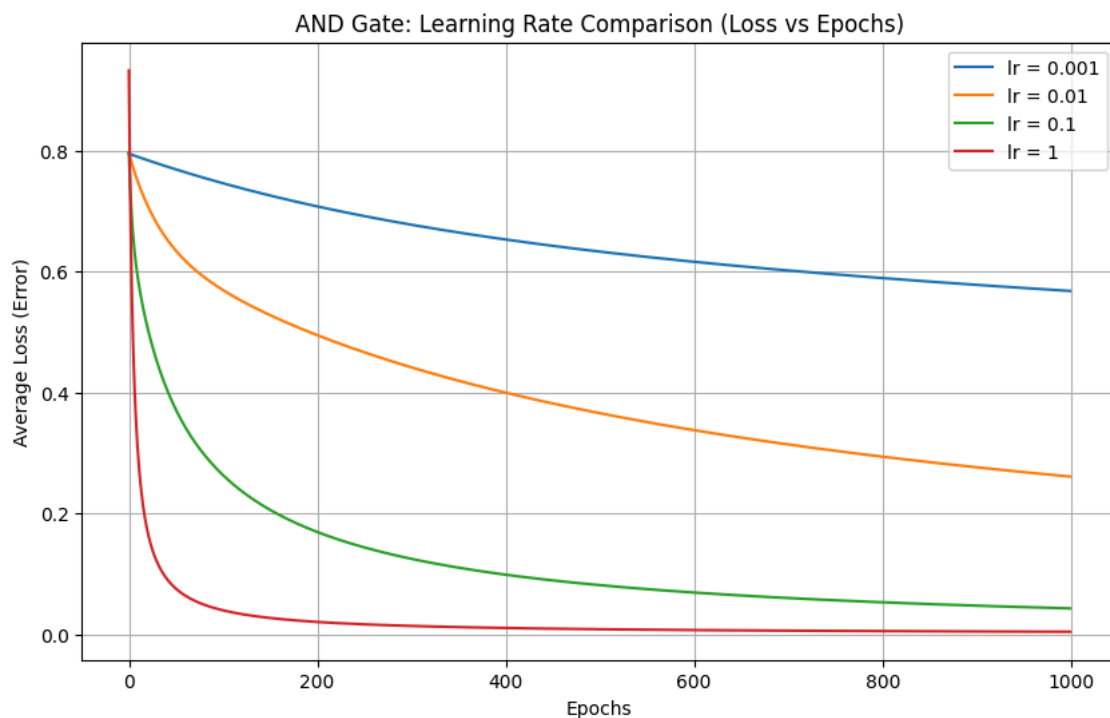
        dz = y_pred - y[i]
        dw = dz * X[i]
        db = dz

        w -= lr * dw
        b -= lr * db

print("Weights:", w)
print("Bias:", b)

```

```
print("Prem Massand,431")
print("Date:05/02/2026")
```



```
Weights: [5.60154076 5.59544869]
Bias: -8.565987677634602
Prem Massand,431
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import numpy as np
import matplotlib.pyplot as plt

X = np.array([
    [0,0],
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    [1,1]
])

y = np.array([0, 1, 1, 1])

def sigmoid(z):
    return 1/(1+np.exp(-z))

learning_rates = [0.001, 0.01, 0.1, 1]

epochs = 1000

plt.figure(figsize=(10, 6))

for lr in learning_rates:

    w = np.array([0.1, 0.2], dtype=float)
    b = 0.3

    loss_list = []

    for epoch in range(epochs):

        total_loss = 0

        for i in range(len(X)):

            z = np.dot(X[i], w) + b
            y_pred = sigmoid(z)

            loss = -(y[i]*np.log(y_pred) + (1-y[i])*np.log(1-y_pred))
            total_loss += loss
```

```

dz = y_pred - y[i]
dw = dz * X[i]
db = dz

w -= lr * dw
b -= lr * db

loss_list.append(total_loss / len(X))

plt.plot(loss_list, label=f"lr = {lr}")

plt.xlabel("Epochs")
plt.ylabel("Average Loss (Error)")
plt.title("OR Gate: Learning Rate Comparison (Loss vs Epochs)")
plt.grid(True)
plt.legend()
plt.show()

lr = 0.1
w = np.array([0.1, 0.2], dtype=float)
b = 0.3

for epoch in range(1000):
    for i in range(len(X)):
        z = np.dot(X[i], w) + b
        y_pred = sigmoid(z)

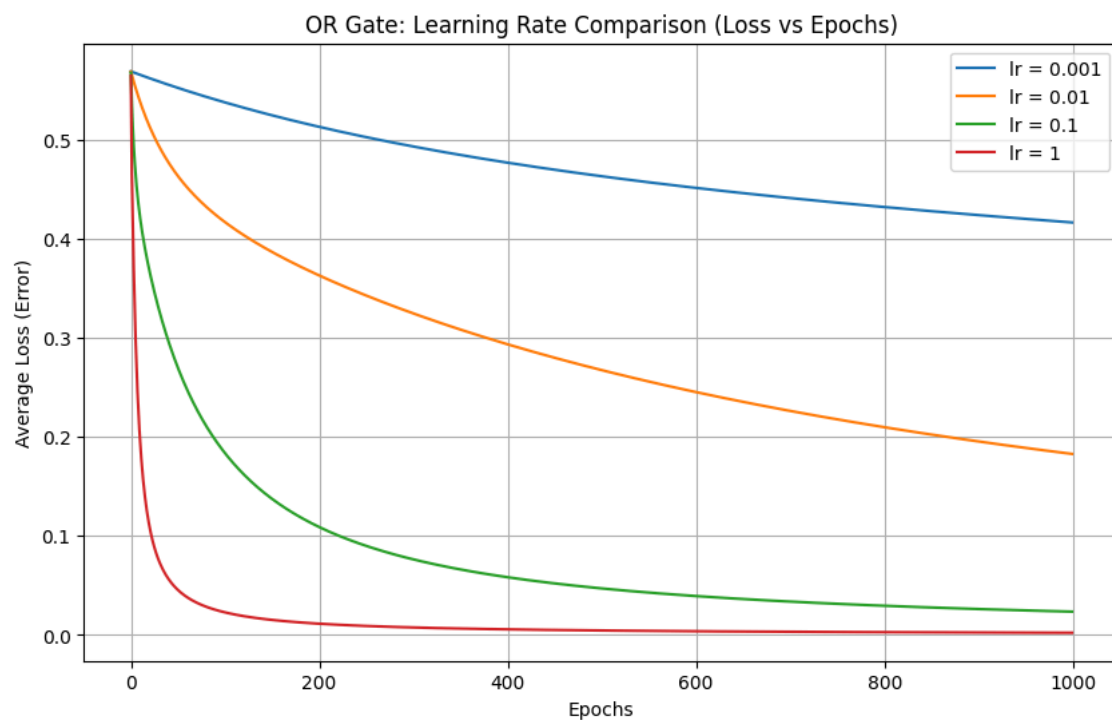
        dz = y_pred - y[i]
        dw = dz * X[i]
        db = dz

        w -= lr * dw
        b -= lr * db

print("Weights:", w)
print("Bias:", b)

print("Prem Massand,431")
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Weights: [6.7832796 6.78742098]
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